esotrace Workshop
Part 3: Command Line Tools

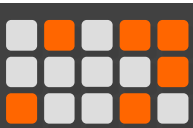
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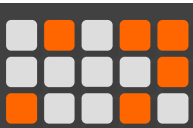
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1. jtracetool

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- **jtracetool – convert logger data**

  java –jar jtracetool.jar [-o <output>] <input_path>

  Input:

tmt, otmt, offline.zip	Logger Data
esotrace files/partial files	Raw esotrace Protocol stream
esotrace directory	Re-convert existing log

  Output:

esotrace directory	Converted log data

  + logger data: time stamps and markers



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Jtracetool: Options

- **Lots of options:**

```
c:\Users\chvo2929\trace_tools_20120829_4.1.0.pre\tools>java -jar jtracetool.jar
usage: jtracetool [options] <otmt_directory|otmt_file|tmt_file>
 -A          in: disable auto detection of format
 -B          in:tmt: disable old block skipping
 -D <arg>    out:eso: set number of digits: 4
 -E          show: logger data and time events
 -e <arg>    in:tmt: select Ethernet channel from input stream: 0
 -F <arg>    in:tmt: set name filter for OTMT file selection: _ETH_
 -G          show: only generated packets
 -I <arg>    in: set input format: auto [tmt eso raw]
 -J <arg>    stream: validate trace levels against given tracing.json file
 -L          show: internal logger events (format specific)
 -l          stream: create logger data and time packets
 -M <arg>    stream: logger time: max packet offset in tmt message: 512
 -m <arg>    stream: restore missing entities in partial stream from given
             .sem file
 -O <arg>    out: set output format: auto [esofile esodir raw]
 -o <arg>    out: select directory/file name/path: null
 -P          show: decoded trace packets
 -R          stream: disable repairing of eso stream if packets are missing
 -r          stream: raw mode. do not process packets only read/write
             bytes
 -S <arg>    out:eso: set split size (in KiB): 10240
 -s          silent operation, print no messages
 -T <arg>    stream: logger time: minimum delta between updates (ms): 500
 -t <arg>    in:tmt: select Ethernet type from input stream: raw [gnlogger,
             utf8, any]
 -U          stream: do not create a new session on detected time warp
 -V          stream: disable stream validation.
 -X          show: only dropped message
 -x          show: raw data of trace packets
 -Y <arg>    show: dump all logger events: 0=off 1=internal 2=all
 -y          stream: recover partial/pre-cut stream
 -Z <arg>    in:tmt: auto detect sample buffer size in MiB

c:\Users\chvo2929\trace_tools_20120829_4.1.0.pre\tools>
```

- **Typical use:**

```
jtracetool (short for java -jar jtracetool.jar)

jtracetool -l -o out.esotrace in.tmt
jtracetool -l -o out.esotrace part1.tmt part2.tmt part3.tmt
jtracetool -l -o - block1.otmt block2.otmt # output is stdout!

-l „el“ = add logger timer stamps and markers to stream
```

The TMT Files must contain a continous session. No gaps allowed!

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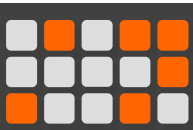
Jtracetool: Options

- Main Options
  - **-o** set output file/directory name
  - **-l** „el“ extract logger time stamps, markers (if available)

- Useful Options:
  - **-s** silent operation = no output -> use in scripts

```
jtracetool -s -o test.esotrace in.esotrace # silent re-pack
```

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Jtracetool: Options

- Format Options
  - -O select output format (default: autodetect)

```
jtracetool -l -O esodir -o out.esotrace in.tmt  # default
jtracetool -l -O esofile -o out.esotrace in.tmt  # single file for transfer
jtracetool -l -O null in.tmt  # no output - quick testing of input
```

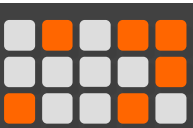
  - -I select input format (default: autodetect)

```
jtracetool -l -I tmt in.tmt  # default
jtracetool -l -I eso in.raw  # file or dir esotrace is detected automatically
```

- Esotrace Output Options:
  - Scheme: dir.esotrace/log_0000.esotrace
  - -D <n> (default: 4) number of digits in container files
  - -S <KiB> (default: 10240) split size of container files in KiB

```
jtracetool -D 5 -S 20480 -o test.esotrace in.esotrace # repack esotrace
```

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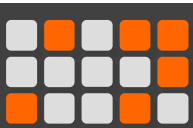


Jtracetool: Options

- Diagnosis
  - -X show dropped message packets
  - -E show logger events: logger time and marker
    - -L show internal logger events (connect state!)
  - -Y 0|1|2 dump all logger events (for tool debug)
    - 0=off 1=internal 2=all
  - -P show all trace packets decoded
    - -G (only with –P) show only generated packets
  - -x show raw data trace packets

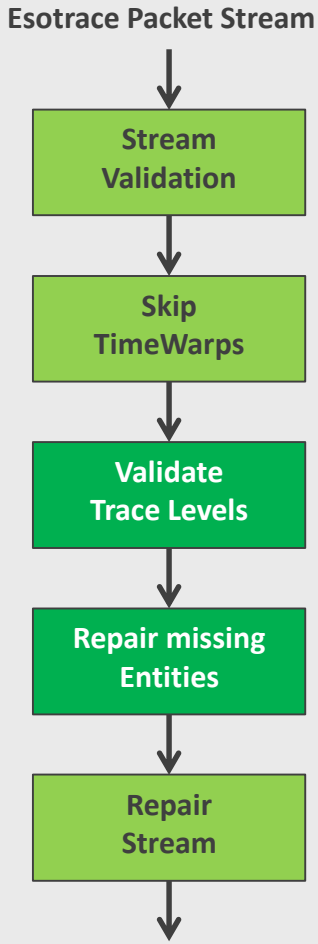
```
jtracetool –LX –O null in.tmt   # quickly scan TMT file for
                                # dropped messages and show logger connects
                                # -> analyse Trace Link Failures
```

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Jtracetool: Options

- Packet Stream Operations
  - -V Disable stream validation
    - Check that all entities are available
  - -U do not create new session on time warp
    - Repair function for broken streams
  - -J tracing.json
    - Validate trace levels against given config
  - -m traceserver.sem
    - Repair pre-cut stream and add missing entities
  - -R Disable stream repair
    - Add EXIT packets if they are missing



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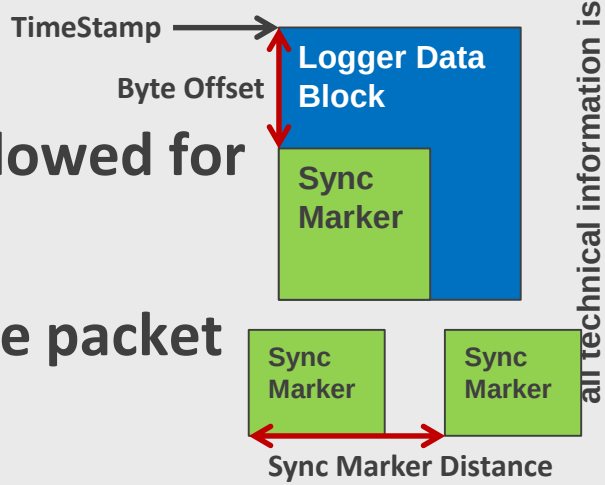
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Jtracetool: Options

- When autodetection fails... Manual Config
  - -A disable auto detection
  - -y Seek SyncMarker in partial stream
    otherwise esotrace stream is assumed to be uncut

```
jtracetool -A -y -o out.esotrace in.tmt   # stream is assumed to be pre-cut
Jtracetool -A -o out.esotrace in.tmt      # stream is assumed to be uncut
```

- Internal Options (LoggerTime Mapping):
  - -M <bytes> max byte offset in packet allowed for associated SyncMarker
  - -T <ms> minimum interval of LoggerTime packet creation



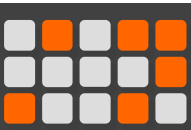
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Jtracetool: Options

- TMT Parameter Detection
  - **-A** disable auto detection (default auto enabled)
  - **-e <num>** set ethernet channel in TMT/OTMT: 0..3
    - Note: TMT Tools use numbering 1..4 !
  - **-F <str>** name filter for OTMT files (default: _ETH_)
    - Matches *_ETH_* use „“ for no filter
  - **-t <type>** select TMT ethernet type of stream
    - Valid values are: raw, gnlogger, utf8, any = default
  - **-Z <bytes>** size of detect buffer for each stream (in MiB) (1)
  - **-B** disable duplicate TMT block skipping (=no repair)

```
jtracetool -A -e 0 -t gnlogger -F _ETH_ dir.otmt  # scan OTMT directory
                                                  # for file *_ETH_*.otmt
                                                  # extract ethernet stream #0
                                                  # (wrong) type GNlogger
```

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Jtracetool: Options

- Raw Mode – Brute Force Repair of badly broken streams
  - „If you have lost all hope...“
  - **-r** enable raw mode
  - only ethernet stream will be unpacked (e.g. from TMT)
  - use **–e** and/or **–t** to select ethernet type

```
jtracetool –r –o dump.raw in.tmt        # dump ethernet stream raw to file
jtracetool –r –e 1 –o dump.raw in.tmt   # dump ethernet stream #1
```

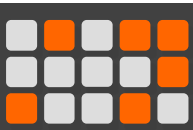
  - Use salvage_esotrace.py Script to recover esotrace packets

```
python27 salvage_esotrace.py –o repair.esotrace dump.raw
```

(more details in Scripts section)

- Sorry: no logger time stamps or markers... ☹

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2. jtracecapture

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- **jtracecapture – „poor man‘s data logger“**
  java –jar jtracecapture.jar [-o <output>] <target_ip>

  Input:
  esotrace Packets from Trace Server via TCP/IP

  Output:
  esotrace Directory with esotrace log

  – Same capture core as found in viewer
  – Retries and re-connects automatically
  – Control server via commands on stdin



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eso TraceTools: 2. jtracecapture

- **Lots of options**

```
c:\Users\chvo2929\trace_tools_20120829_4.1.0.pre\tools>java -jar jtracecapture.jar
usage: jtracecapture [options] <hostname>
 -A <arg>    auto quit after a delay (ms)
 -a <arg>    auto drop connection after a delay (ms)
 -B          hide begin/end session title
 -b <arg>    use backend for output
 -C          disable text cropping
 -c <arg>    maximum connect duration (ms)
 -D          hide row type description
 -d <arg>    report data volume interval in ms: 2000
 -E <arg>    using emstable manager: null
 -e <arg>    emstable archive file: emstables.zip
 -F <arg>    using file download manager: null
 -f <arg>    write text log to file
 -G <arg>    statistics name level 0=only summary 1=process, 2..n channels
 -g          generate statistics (only summary)
 -i <arg>    idle time out in s: 5
 -K <arg>    delay between connect retries (ms): 500
 -k <arg>    number of connect retries (0=infinite): 20
 -L <arg>    write a log file
 -l          add logger timestamps
 -m          multi file output (per session)
 -n          do not add logger time to sync markers
 -o <arg>    binary output directory/file: esotrace_#yyyyMMdd#
 -P          print decoded trace packets
 -p <arg>    host port: 21002
 -r <arg>    enforce pre connect delay (ms): 0
 -S          silent operation. no output
 -s          stay connected and ignore idle time outs
 -T          use tabs to separate text fields
 -t <arg>    reach host time out (s) (0=infinite): 120
 -X          print dropped messages
 -x          do not use external ping command to detect target
 -Y          enable debug mode

c:\Users\chvo2929\trace_tools_20120829_4.1.0.pre\tools>
```

- **Typical usage:**

```
jtracecapture (short for java -jar jtracecapture.jar)

jtracecapture target
jtracecapture -o test.esotrace 172.16.250.248
```

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Jtracecapture: Options

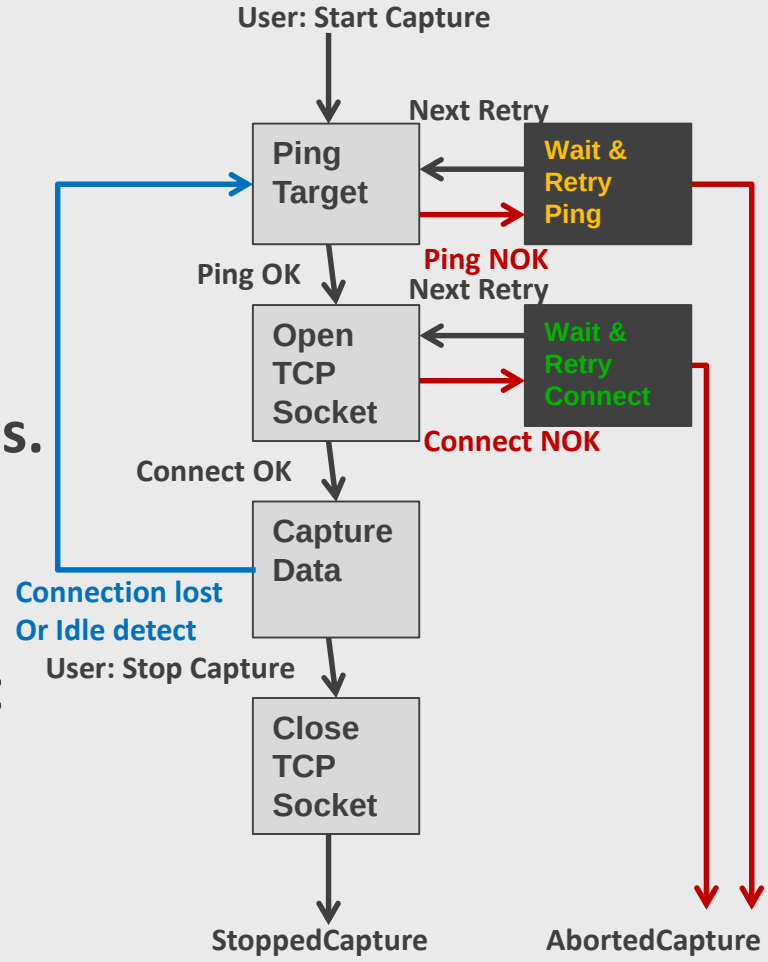
- Main Options
  - -o set output esotrace directory name
  - -p <num> set capture port (default: 21002)
  - -l add „logger“ timestamps = PC time
    - -n only markers no time stamps

- U
```
jtracecapture -S target  # capture without lots of console output
```
  - -S silent operation = no output -> use in scripts
  - -d <ms> update interval of online status messages
    Default: 2000 ms

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Jtracecapture: Options

- Connection State Machine
  - **-t <s>** maximum time for ping retries. default: 120 s
    0 = wait forever
  - **-k <n>** number of connect retries.
    Default: 20
    0 = wait forever
  - **-K <ms>** delay between connect retries.
    Default: 500 ms



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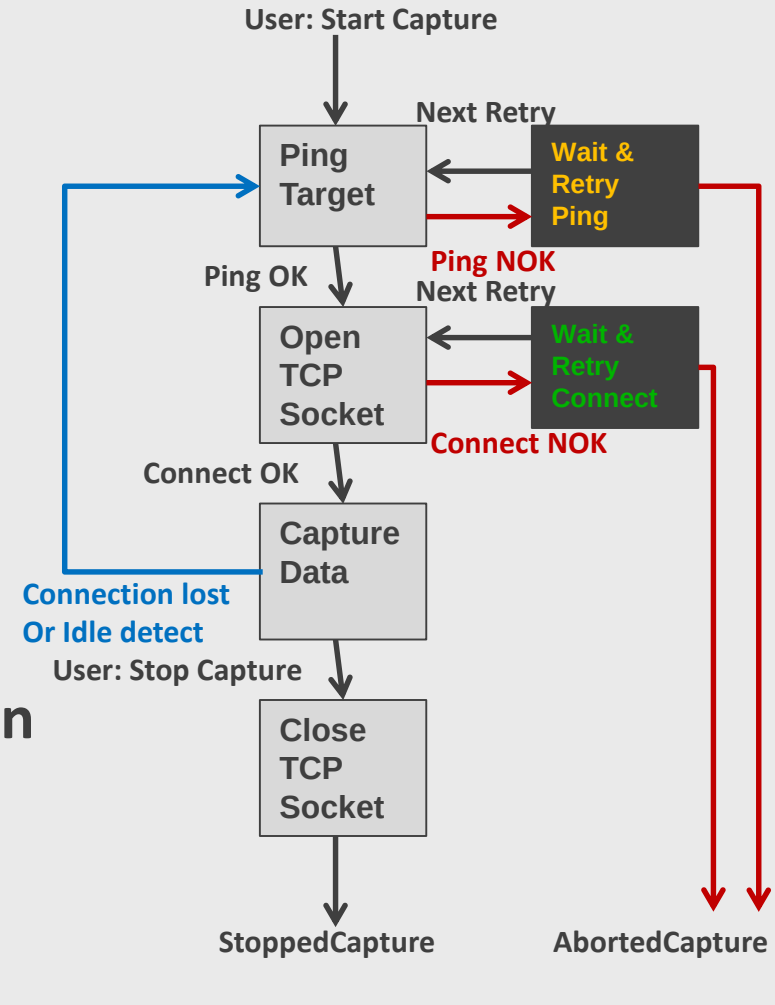
```
jtracecapture -t 0 -k 0 target  # setup suitable for non-stop tests
```

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Jtracecapture: Options

- Connection State Machine (cont.)
  - **-i <s>** idle time out
    Default: 5s
  - **-s** stay connected and ignore idle time o
    (Debug only)
  - **-r <ms>** enforce a delay between ping and connect.
    Simulates late connects of a logger



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Jtracecapture: Options

- Special Capture Modes for Debug and Testing
  - -c <ms> maximum connection duration
    - Terminate connection after given interval
    - Quit application
  - -a <ms> auto disconnect after interval
    - Terminate connection
    - Retry connect
  - -A <ms> auto quit application after interval
    - Terminate tool
    - No matter what connection state was established

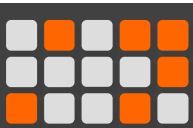
```
jtracecapture -c 20000 target  # let the capture run for 20s
jtracecapture -a 10000 target  # test multiple connects with 10s each
```

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Jtracecapture: Options

- Direct text log export
  - -f output.log write a text log file
  - Lots of Text Export Options:
    - -B, -C, -D, -T, -m, -e
    - See same options in jtracedump Tool!

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Jtracecapture: Options

- Diagnosis
  - -X show dropped messages
  - -P print received esotrace packets
  - -x do not use external „ping“ command
    - Use internal (only if external fails)
  - -Y enable tool debug mode
    - Slower, but WatchDog whole capture core
    - -L test.log write debug log file of tool

- SDK Support
  - -b <backend.class.name>
    - Attach an own backend to receive captured data
    - Backend class must be in Java CLASSPATH

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Jtracecapture: Options

- Generate Profiling Output
  - Extract Statistics for Processes, Channels
  - Count number of log messages, transferred bytes

-g enable generation of statistics files in directory „capture.data“

-G <num> generate individual statistics files for channel namess up to <num> levels

0 = only process names

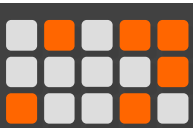
1 = first level of channels

...

999 = all channel levels ;)

- Use **trace_datvis.py** script to visualize data (see Scripts)

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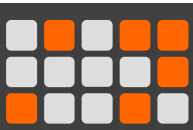


3. jtracedump

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- **jtracedump – convert esotrace stream to readable logs**
  java –jar jtracedump.jar [-o <output.log>] <input.esotrace/>

  Input:

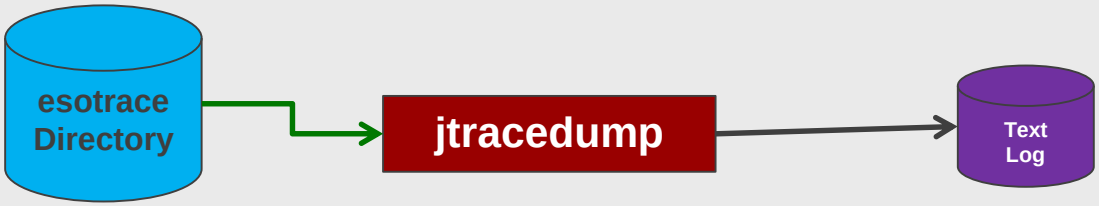
  esotrace Directory with esotrace log

  Output:

  *.log (UTF8)-Textfile with Log

  Extra Features:

  – Generate statistics -> Plot Load Graphs



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eso TraceTools: 3. jtracedump

- **Lots of options**

```
usage: jtracedump [options] <esotrace dir>
 -a          abbreviate output: one line per entry
 -b <arg>    feed output to given backend class
 -B          hide begin/end session title
 -C          disable text cropping
 -D          hide row type description
 -E <arg>    use ems table manager
 -e <arg>    emstable archive file: emstables.zip
 -f <arg>    use file download manager
 -g          generate statistics
 -G <arg>    statistics name level 0=only summary 1=process, 2..n channels
 -i          show statistics
 -J <arg>    check levels of tracing.json file
 -l          add logger timestamps
 -M <arg>    export messages of given type number raw into file
 -m          multi file output (per session)
 -n          use null exporter for output
 -o <arg>    text/data file output path
 -s <arg>    dump only one session
 -S          be silent, no output
 -T          use tabs to separate text fields

C:\Users\chvo2929\trace_tools_20120829_4.1.0.pre\tools>
```

- **Typical usage:**

```
jtracedump (short for java -jar jtracedump.jar)

jtracedump -o test.log dir.esotrace  # must be a esotrace directory!

# you can „fake" a directory if you only have a single esotrace file
mkdir my.esotrace
mv in.esotrace my.esotrace/log_0000.esotrace
jtracedump -o test.log my.esotrace
```

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Jtracedump: Options

- Main Options
  - -o set output esotrace directory name
  - -l convert logger time stamps and add logger time column

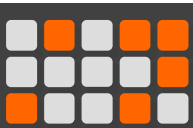
- Useful Options:

```
jtracedump -S in.esotrace  # convert silently
```

  - -S silent operation = no output -> use in scripts

- Multi Session
  - -m create a new file for each session file_0.log ...
- Single Sesion
  - -s <num> pick only a single session

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Jtracedump: Options

- Text output formatting
  - **-a** print only first line of log message (default: multi lines)
  - **-B** do not print session header/footer
  - **-C** do not crop text in columns
  - **-D** do not print column descriptions/headers
  - **-T** use TABs to separate columns
    - Useful for applications with TSV import



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Jtracedump: Options

- COMM Message Decoding
  - RPC Calls are recorded as blobs in the trace stream
  - Need decoding tables to reconstruct plain text calls
  - ems_tables.zip from Target in /eso
  - -e <ems_tables.zip> enable COMM decoding

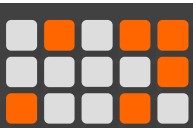
- Raw Export of Custom Message Types
  - Useful for embedded trace streams
  - -M <msg_type_num> e.g. –M 130 Navi TPEG
  - -o <prefix> give prefix
  - Creates files for each session:
    - <prefix>_<session_num>.dat

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Jtracedump: Options

- Diagnosis
  - -n null output
    - Just to check if import processing works
- Check Trace Levels
  - -J tracing.json
  - Validate that all generated messages match the filter profile given in the tracing.json
- SDK Support
  - -b <backend.class.name>
    - Attach an own backend to receive log data
    - Backend class must be in Java CLASSPATH

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Jtracedump: Options

- Generate Profiling Output
  - Extract Statistics for Processes, Channels
  - Count number of log messages, transferred bytes

-g enable generation of statistics files in directory „capture.data“

-G <num> generate individual statistics files for channel namess up to <num> levels

0 = only process names

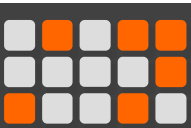
1 = first level of channels

...

999 = all channel levels ;)

- Use **trace_datvis.py** script to visualize data (see Scripts)

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Coffee Break

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4. jtracelogger

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eso TraceTools: jtracelogger

- **Predecessor of jtracecapture: „capture with a core“**
- **Options:**

```
$ java -jar jtracelogger.jar
usage: jtracelogger [options] <hostname>
 -b          output a binary log
 -c          use console output
 -C          disable text cropping
 -D <arg>    set number of digits: 4
 -f <arg>    human readable log file name: eso_#yyyyMMdd_HHmmss#.log
 -I          ignore filter levels of entities in stream
 -i          use interactive console
 -l          output a human readable log
 -n          use null backend for output
 -o <arg>    binary output directory/file: esotrace_#yyyyMMdd_HHmmss#
 -P          dump trace packets
 -p <arg>    set trace port: 21002
 -s          be silent
 -S <arg>    set split size (in KiB): 1024
 -T          use tabs to separate text fields
```

very similar to the ones in jtracecapture

-

```
jtracelogger (short for java -jar jtracelogger.jar)

jtracelogger target
jtracelogger -o test.esotrace 172.16.250.248
```

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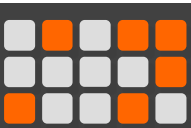
eso TraceTools: jtracelogger

- **Interactive Mode**

```
jtracelogger -i 172.16.250.248
```

- **Press <Enter> to enter/leave command shell**
  - **? = show help**
  - **ls = show all channels, callbacks, threads**
  - **channel <name|id> [level] = get/set level**
  - **thread <name|id> [level] = get/set level**
  - **wait thread|channel <name|id> = wait until channel is here**
  - **call <name|id> [args] = execute callback**
  - **mute/unmute = show/hide incoming messages**
  - **quit = shutdown logger**

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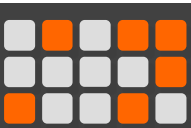
eso TraceTools: jtracelogger

- You can pipe in commands via stdin (for scripting)

```
echo -e „mute\nwait channel my.channel\ncall my.channel\nquit\n“ | \
 jtracelogger -i 172.16.250.248
```

- Note:
  - In contrast to jtracecapture: handles only one session
  - No re-connect, no connect state machine
  - Reads optional config file `config/tracing.json`
    - Section: „Logger“
  - Allows to attach backends via config or with -b

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5. jtracecmd

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eso TraceTools: jtracecmd

- Tool to control the server: change levels or trigger call backs
- Options:

```
usage: jtracecmd [options] <cmd1> [args1] [+ <cmd2> [args2] ...]
 -H <arg>   target host name. default: target
 -P <arg>   prefix to be prepended to all entity paths. default:
            traceserver.
 -p <arg>   target port. default: 21002
 -s         silent operation. no log output
 -t <arg>   timeout for commands in ms. default: 5000

Available Commands:

call
        <callback_name> [callback_arg]
                Execute a trace callback on the target

sync
                Wait for next sync marker from target

begin_capture | bc
        <channel|'any'> [thread|'any']
                Start capturing messages of the given channel and/or thread

end_capture | ec
        [wait in s]
                End capturing messages. Allows you to wait a bit for messages first.

pring_msgs | pm
                Print the last captured messages

call_capture | cc
        <channel|'any'> <thread|'any'> <capture time in s> <callback> [arg]
                Call a callback and log resulting trace messages

channel
        <channel_name> [level]
                Show or set filter level of channel

thread
        <thread_name> [level]
                Show or set filter level of thread

probe
        <probe_name> [level]
                Show or set filter level of probe

channel_tree
        <channel_name> [level]
                Show or set filter level of channel

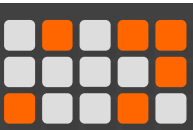
delay
        <time in s>
                Delay execution of next command
```

Typical Usage:

```
jtracecmd (short for java -jar jtracecmd.jar)

jtracecmd -H <hostname> call myapp.mycallback
jtracecmd -H <hostname> channel_tree myapp.mychannel
```

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eso TraceTools: jtracecmd Options

- **Options set global parameters before commands**

```
jtracecmd –H <hostname> -p <port> -P traceserver.myapp <command> ….
```

- **Options**
  - **-H <hostname> = set target hostname or IP address**
  - **-p <port> = set trace server port (default: 21002)**
  - **-P <path> = set default path prefix prepended to all channels, threads, or callback paths given in commands**
  - **-s = silent operation. No extra output**
  - **-t <ms> = set a global timeout in ms for command processing**

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eso TraceTools: jtracecmd Commands

- Either one command or a chain of commands can be given:

```
jtracecmd <command1> [args1] [ + <command2> [args2] …]
```

- Commands
  - **call:** trigger a callback
  - **sync**: wait for a sync marker
  - **begin_capture |bc:** start capturing log messages
  - **end_capture |ec**: stop capturing log messages
  - **print_msgs | pm**: print captured messages
  - **call_capture |cc**: trigger a callback and record log messages as a reply
  - **channel, thread, probe, thread_tree**: show or set filter level of entity
  - **delay**: delay execution of next commands

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eso TraceTools: jtracecmd Commands

- **Call Command: <callback_name> [callback_arg]**

```
jtracecmd call myapp.mycallback test_param
```

- **Capture: begin, end, print**

```
jtracecmd bc <channel>|any <thread>|any + delay 5 + ec + pm
```

- **Combined: call and then capture result**

```
jtracecmd cc <channel>|any <thread>|any <capture time in s> <callback> [arg]
```

- **Query the filter level of a channel, thread, channel tree**

```
jtracecmd channel|thread|channel_tree <entity_path_name>
```

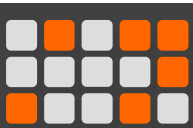
- **Set the filter level of a channel, thread, channel tree**

```
jtracecmd channel|thread|channel_tree <entity_path_name> <new filter level>
```

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6. jtracecheck

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eso TraceTools: jtracecheck

- **Logger.json file validator and extractor tool**
- **Options:**

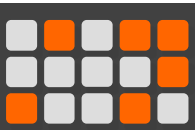
```
Z:\work\perforce\mib_dev_main\build_linux64\java\bin>java -jar tools_trace_jtracecheck.jar
jtracecheck 4.1.5 2013-09-10CEST 10:56
Usage: jtracecheck [options] <command> <argument>
Commands:
  ts | timestamp  <string> ...                check time stamp format
  mi | metainfo   <logger.json> ...           parse a metainfo file
  mf | matchfile  <logger.json> <path> ...    check if a path matches the data file naming
  lf | listfiles  <logger.json> ...           list all associated data files
  ls | listsessions <logger.json> ...         parse and display session data
  wb | writebytes <logger.json> ...           write sessions bytes to files
  wp | writepackets <logger.json> ...         write sessions packets to files
usage: Options
 -v   be more verbose
```

- **Typical Usage:**

```
jtracecheck (short for java –jar jtracecheck.jar)

jtracecheck –v mi <logger.json>
Jtracecheck –v lf <logger.json>
```

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eso TraceTools: jtracecheck Commands

- Commands:
  - **ts | timestamp** = check if given string is valid timestamp
  - **mi | metainfo** = read a logger.json file and check if the json syntax and semantics are correct
  - **mf |matchfile** = use the data file pattern from the logger.json file and try to match it to files in the current directory
  - **lf |listfiles** = list all esotrace data files that are mentioned in the logger.json file
  - **ls |listsessions** = list the esotrace sessions stored in the data files and show their locations inside the files
  - **wb |writebytes** = extract all the mentioned sessions in a logger.json file into individual data files
  - **wp |writepackets** = similar to wb but also check that valid esotrace packets are found in the sessions

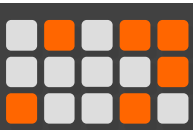
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7. jtraceconv

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eso TraceTools: jtraceconv

- File format converter. New replacement for jtracetool
- Options:

```
290@eso2978: ~ [63] $ jtraceconv
: jtraceconv [options] <input> [input ...] <output>
      enable debug output
arg>  set exporter options: key=value;key=value...
arg>  select exporter
arg>  extra help: use 'importer' or 'exporter' to show available
      plugins or use 'i:name' or 'e:name' for options
arg>  set importer options: key=value;key=value...
arg>  select importer
      silent operation, print no messages
arg>  repair names during import with given .sem file
arg>  temporary esotrace directory for impex operation:
      temp.esotrace
290@eso2978: ~ [64] $
```

- Typical Usage:

```
jtraceconv (short for java -jar jtraceconv.jar)

jtraceconv <input files…> <output file>
Jtraceconv input.json output.esotrace
```

  - Creates esotrace dirs from single or multiple input files
  - Suitable for *.json to esotrace conversion

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8. tracebatch

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eso TraceTools: tracebatch

- **Process a large set of trace files in a single run**
- **Options:**

```
chvo2929@eso2978: ~ [65] $ tracebatch.py
Usage: tracebatch.py [options] <dir> ...

Options:
  -h, --help            show this help message and exit
  -v, --verbose         show output during processing
  -n, --dry-run         to not perform actual processing
  -o OUTPUT_DIR, --output-dir=OUTPUT_DIR
                        output directory for commands generating output
  -O OUTPUT_FORMAT, --output-format=OUTPUT_FORMAT
                        select output format for converter commands
  -d DAY_OLD, --day-old=DAY_OLD
                        only examime dirs not older than given days
  -F, --skip-offline    skip offline traces during walk
  -N, --skip-normal     skip normal traces during walk
  -t FILE_TYPE, --file-type=FILE_TYPE
                        only pick input files of given type
version <dev> <nodate>

Commands:
ls      list all found traces
can_conv        check if traces can be converted
test_conv       test conversion into temp dir
json_check      check all json files
cat     print all traces as text
grep    search for given expression
chvo2929@eso2978: ~ [66] $
```

- **Typical Usage:**

```
python tracebatch.py <command> <input dirs>

tracebatch.py grep ERROR ~/ADERA  # scan whole directory for string
tracebatch.py cat ~/dir | egrep ERROR # cat all logs found there
```

  - **Detects various formats and can call other tools to process them**

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9. Scripts

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Scripts Overview

- **esotrace Helpers**
  - **pmtrace.py** – recover unsent trace message in a process from a core dump
  - **salvage_esotrace.py** – recover badly damaged esotrace streams
  - **tmascii2esotrace.py** – recover esotrace streams exported in ASCII format
  - **strip_gnlog.py** – remove a GNLog file container around an esotrace stream
  - **xtmt2esotrace.py** – convert BluePiraT2 XTMT Files to logger.json and esotrace data files

- **Serial Tracing Helpers**
  - **power_capture.py** – perform power on/off test and record serial streams
  - **serhex2txt.py** – recover serial traces that were exported in TM hex dump format

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esotrace Helpers

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pmtrace.py – post mortem tracing

- Extract esotrace information from a core dump

```
pmtrace.py <options> dump.core

 This tool extracts eso Trace Messages from a Core Dump

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 (The "readelf" utility is required as an external tool!)
 (Do not forget to include data segments in your core dump.
  On QNX: do not use dumper with -m switch)

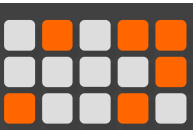
 -v                  be more verbose
 -n                  don't try to find and decode entity pools (-> no names)
 -o <file>           write messages to file
 -b                  write binary trace file (default: readable text trace)
 -s                  use ipl_string decoder for FW 3.5.3 (default: const char *)
 -r <readelf>        set the readelf binary (default: readelf)
```

- Create and store a core dump _with_ data segments (QNX: dumper –m)
- Run tool with:

```
python27 pmtrace.py -b -o out.esotrace my.core
```

- -n = may find more but ignores channel names
- -b –o <name> = esotrace output
- -o <name> = text output
- -r <bin> = set the path of the `readelf` binary (is contained in QNX cross tool chain)

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salvage_esotrace.py – Recover badly damaged streams

- Reconstruct an esotrace stream that can contain invalid packet structures (i.e. Internal damage not pre- or post-cut files!)
- Call tool with:

```
python27 salvage_esotrace.py -o output.esotrace input.raw

# pipe operation does not require intermediate file creation:
jtracetool -r -o - *.otmt | python27 salvage_esotrace.py -o out.esotrace -
```

- More options:

```
$ python salvage_esotrace.py
Usage: salvage_esotrace.py [options] <file> ...

Options:
  -h, --help            show this help message and exit
  -o OUTPUT, --output=OUTPUT
                        write salvaged packets to file
  -v, --verbose         show packets
  -V, --verbose-contents
                        show contents of packets
  -s SKIP_SHOW, --skip-show=SKIP_SHOW
  -g, --generate-skip-pkt
                        generate a logger data packet for each skipped range
  -D, --generate-esodir
                        generate a esotrace directory as output
  -H, --generate-hash   generate hash for all messages
  -c, --check-hash      check for hash duplicates
  -E, --check-exit      try to salvage EXIT packets, too
```

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tmascii2esotrace.py – re-binarize traces

- If a trace was exported in the wrong format (here Telemotive ASCII) instead of esotrace RAW binary
- Call tool with:

```
python27 tm2ascii2esotrace.py [-e <ethernet_channel>] [-o <out.esotrace>] <in.txt> …

ethernet_channel: 0...3  logger channel used for recording the esotrace
Otherwise auto detection uses the first channel found
```

- Note:
  - the tool automatically adds Logger Time Stamps as they are found in the txt file, too
- Options:
  - Use –n to disable automatic logger time creation
  - Use –r to enable raw mode and do not decode any esotrace packets. This is useful to decode non-esotrace streams from hex dumps

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strip_gnlog.py – remove a GNLog file container

- If an esotrace was exported in the GNLog file format then this tool can remove the GNLog container and extracts the original esotrace data
- Call tool with:

```
python27 strip_gnlog.py input.13aa [more_input] -o output
```

- Options:
  - -v = be verbose
  - -o <ouput file> = set output file. Use **–** or no **–o** switch for stdout
  - -f = force output to tty
  - -s = show skipped parts of input file (i.e. the GNLog container)
  - -c <channel> = pick the channel to extract the esotrace from (default: autodetect)
    - **0, a, c**: typical data channels
  - -b <bufsize> = buffer size for I/O operation (default: 4096)
- You can directly pipe the output of strip_gnlog.py into salvage_esotrace.py or jtracetool!

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xtmt2esotrace.py – extract esotrace and logger.json from xtmt

- If an esotrace is embedded in an BluPiraT2 XTMT file then this script can extract the esotrace data stream and also creates a logger.json file with marker and logger time information.
- Call tool with:

```
python27 xtmt2esotrace.py <input.xtmt> [more.xtmt]
```

- Options:
  - -v = be verbose
  - -e <channel> = select ethernet channel to be extracted (default: take first encountered)
  - -r <res_ms> = generate data events in given resolution (default: 1ms)
  - -g <gap_ms> = automatically detect a new session if there was no data incoming
- The tool will automatically deduce the output file name from the given input
- Pass in all xtmt files that belong to a single part (e.g. *_#01). Even if they are spanned across days. The script will automatically assign the files to parts.
- You can also extract XTMT files embedded inside Offline.zip files with this tool

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Serial Tracing Helpers

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power_capture.py – serial trace tool

- This tool records serial traces from one or more serial adapters on a system and writes log files with time stamping (similar to capture mode in TeraTerm or PuTTy)
- Can control a power supply unit via serial and schedule power on and cycles (We support PeakTech PSU from Reichelt)
- Can listen on incoming serial expressions and respond with serial output (-> send commands to target on serial console)
- Requires: PySerial Module: `http://pyserial.sourceforge.net/`
- Call tool with:

```
python27 power_capture.py <power_serial>|none <console_serial1> ...

# windows
python27 power_capture.py COM1 COM8 COM9
# linux
python27 power_capture.py none /dev/ttyUSB0
```

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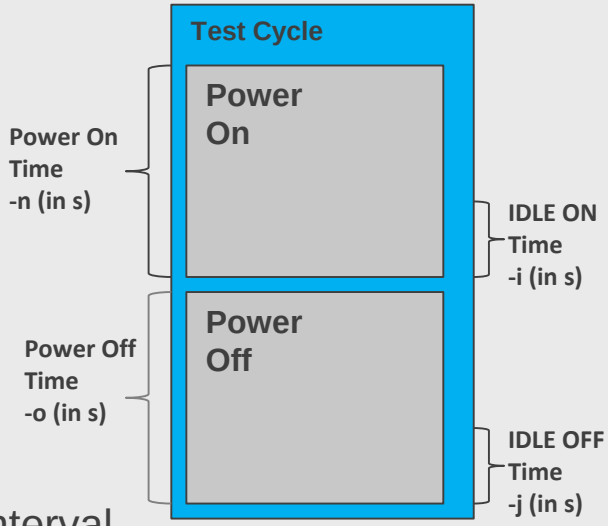
power_capture.py – serial trace tool

- All options:

```
$ python power_capture.py
Usage: power_capture.py [options] <power_serial_port|none> <capture_serial_port> [more_

Options:
  -h, --help            show this help message and exit
  -n POWER_ON_TIME, --power-on-time=POWER_ON_TIME
                        power on time in sec
  -o POWER_OFF_TIME, --power-off-time=POWER_OFF_TIME
                        power off time in sec
  -i IDLE_ON_TIME, --idle-on-time=IDLE_ON_TIME
                        max time in sec to be idle
  -j IDLE_OFF_TIME, --idle-off-time=IDLE_OFF_TIME
                        max time in sec to be idle
  -c CYCLES, --cycles=CYCLES
                        number of test cycles. 0=infinite
  -l LOG_PREFIX, --log-prefix=LOG_PREFIX
                        write log files with given prefix
  -L LOG_SPECIAL, --log-special=LOG_SPECIAL
                        prefix for special log files
  -s, --capture-log     also write console to log
  -q, --quit-on-idle    quit program if an idle occurred
  -C CONTROL_SCRIPTS, --control-scripts=CONTROL_SCRIPTS
                        enable control scripts: port:file,port:file

Available COM ports:
('COM8', 'Prolific USB-to-Serial Comm Port (COM8)', 'USB\\VID_067B&PID_2303&REV_0400')
```



- IDLE means: no incoming serial traces in a given time interval
- -l log = write log files for each serial adapter: log_COM8.log, log_COM9.log (use %d in name to create a new log file for each test cycle)
- -L log = only write into this log file if an IDLE condition occurred. Naming similar to –l option
- -s also add all console messages to the log files
- -C <file_name> use given script to match incoming data and send responses

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power_capture.py – control script

- See „power_capture_example_script“ in scripts directory

```
# match for 'HB' and issue pidin command on console
HB.*System                      # match line: non-indented, regular expression
    pidin\n                     # response lines: indented (spaces/tabs)
    echo "Num %d"\n             # %d is replaced with a number (incremented per match)
    \p                          # special code for “leave power cycle”
```

- Without instrumenting target code you can react on conditions reported by serial traces:
  - Record system info (pidin) or kernel traces
  - Store away files for each occurrance (mv + %d!)
- Real-life example:

```
Starting bundle 88
          on -p100 tracelogger -w -b 512 -k 3000 -F2 -r -f /tmp/tracelogger.kev &\n

FATAL setNotification took longer than 500
          slay -s SIGTERM tracelogger\n
          pidin th > /mnt/ota/pidin.th.%d\n
          mv /tmp/tracelogger.kev /mnt/ota/nc%d.kev\n
          sync\n
```

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serhex2txt.py – recover serial traces exported as hex dump

- If a serial trace is delivered as a Telemotive Hexdump File then this tool can recover the original text and re-add the time stamps
- Call tool with:

```
python27 serhex2txt.py < input.txt > output.txt
```

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Thank You!

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